

I225E Statistical Signal Processing

6. Spectral Analysis I

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Periodic Processes

- Consider a periodic process of the form

$$x(t) = A \cos(2\pi\omega t + \varphi)$$

where:

A is called the amplitude;

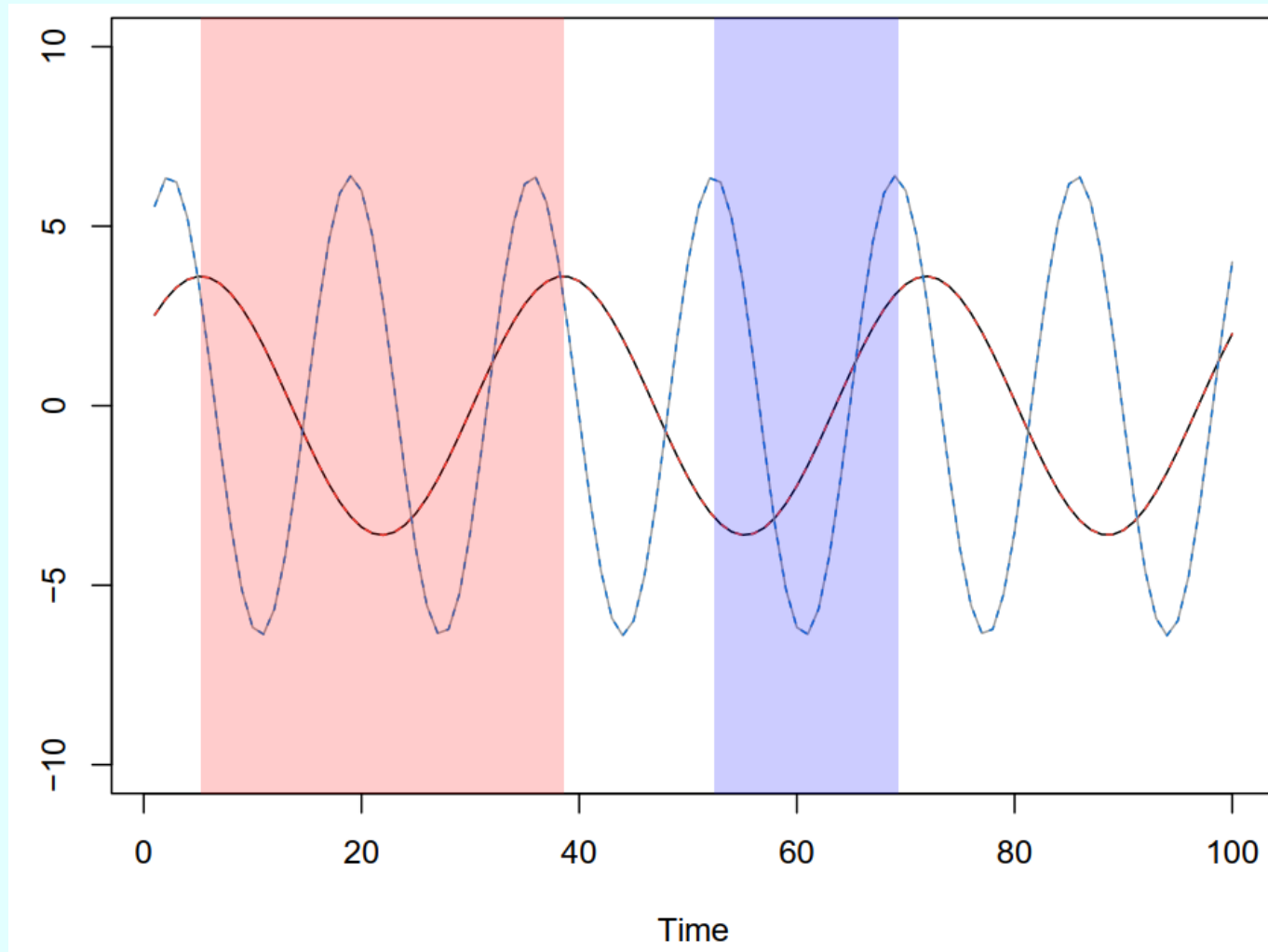
φ the phase of the process;

ω is called frequency of the process; and

$T = 1/\omega$ is called the period or cycle.

- As t varies from 0 to $1/\omega$, note that the process goes through one complete cycle.

Periodic Processes



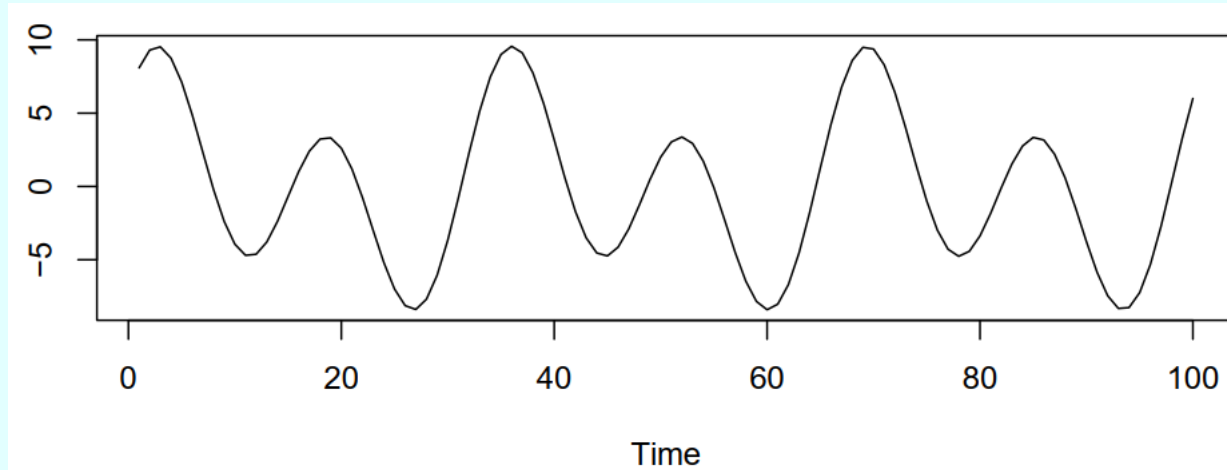
Example: General Mixtures

- We can mix together a total of p periodic processes, which can be expressed as follows:

$$x(t) = \sum_{i=1}^p (U_{j_1} \cos(2\pi\omega_j t) + U_{j_2} \sin(2\pi\omega_j t))$$

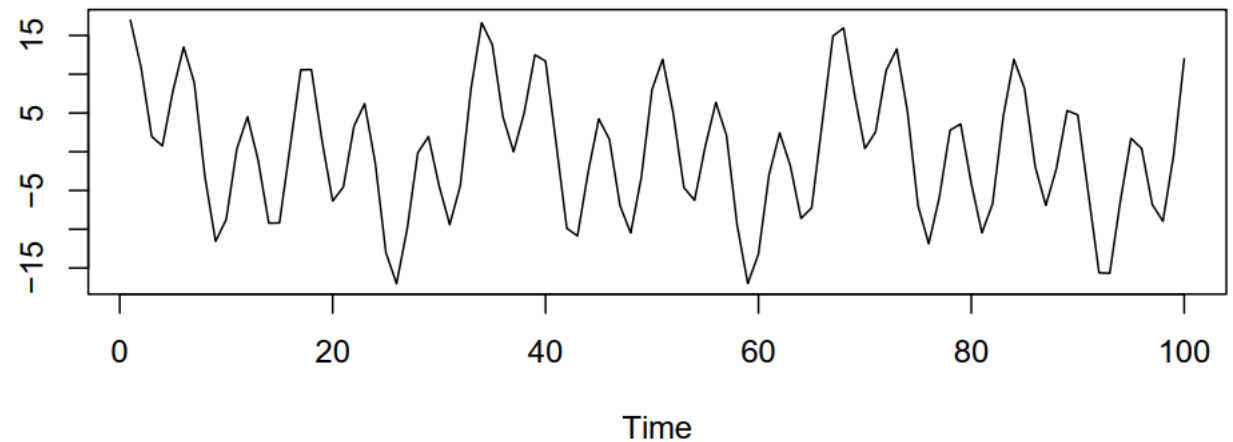
for $U_{j_1}, U_{j_2}, j = 1, \dots, p$ all uncorrelated random variables with mean zero, where U_{j_1}, U_{j_2} have variance σ_j^2 .

Example: General Mixtures



$$p = 2$$

$$p = 3$$



Fourier Decomposition

- A Fourier series is a way to represent a periodic function as a sum of simple sine and cosine waves (or complex exponentials) with different frequencies and amplitudes.
- For a signal $x(t)$ with a period T , its Fourier series representation is given by:

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

where $\omega_0 = \frac{2\pi}{T}$ is the fundamental frequency, and the coefficients a_0, a_n, b_n are calculated using integrals of the function over one period.

Fourier Decomposition

- Alternatively, using complex exponentials, the Fourier series can be written as:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

where the complex Fourier coefficients c_n are given by:

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt$$

- The Fourier series provides a discrete frequency spectrum of the periodic function, consisting of the amplitudes and phases of the fundamental frequency and its integer multiples (harmonics).

Fourier Series to Fourier Transform

- The Fourier transform extends the concept of frequency analysis to non-periodic functions defined over an infinite interval $(-\infty, \infty)$. Consider as the period T of a function approaches infinity.
 - The fundamental frequency becomes infinitesimally small (ω_0 approaches zero)
 - The discrete spectrum becomes continuous ($n\omega_0$ effectively merge into a continuous frequency variable ω)
 - The sum becomes an integral.
- Let's define a function $X(\omega)$ as a scaled version of complex Fourier coefficients c_n .

$$X(n\omega_0) = Tc_n = \int_{-T/2}^{T/2} x(t)e^{-jn\omega_0 t} dt$$

Fourier Series to Fourier Transform

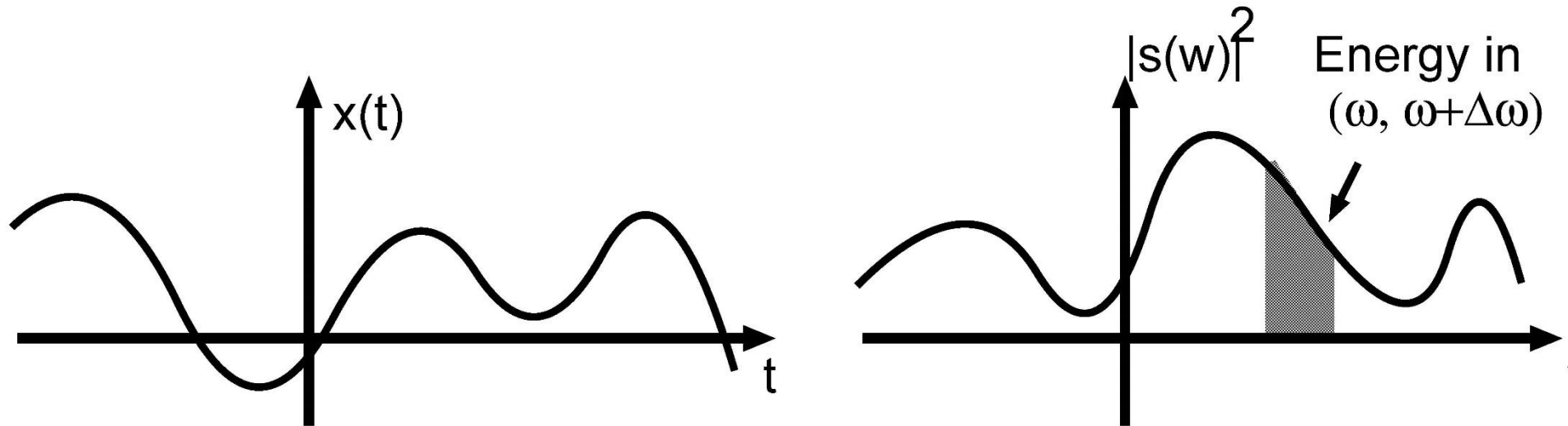
- As $T \rightarrow \infty$, $\omega_0 \rightarrow d\omega$ and $n\omega_0 \rightarrow \omega$. The summation in the synthesis equation of the Fourier series becomes an integral:

$$\begin{aligned}x(t) &= \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \\&= \sum_{n=-\infty}^{\infty} \frac{1}{T} X(n\omega_0) e^{jn\omega_0 t} \\&\approx \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega\end{aligned}$$

- The analysis equation for the Fourier series coefficients becomes the Fourier transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Spectral Representation



For deterministic signal $x(t)$, its spectrum is computed by Fourier transform

$$s(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$

Fourier transforms

- Fourier and inverse Fourier transformations:

$$x(t) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{i\omega t} s(\omega),$$
$$s(\omega) = \int_{-\infty}^{+\infty} dt e^{-i\omega t} x(t).$$

- A Fourier and a subsequent inverse Fourier transform form an identity mapping:

$$\begin{aligned} x(t) &= \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{i\omega t} s(\omega) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{i\omega t} \int_{-\infty}^{+\infty} dt' e^{-i\omega t'} x(t') \\ &= \int_{-\infty}^{+\infty} dt' \delta(t - t') x(t') = x(t). \end{aligned}$$

Parseval's theorem

- The power in the temporal domain is equal to the power in the frequency domain:

$$\begin{aligned}\int_{-\infty}^{+\infty} dt |x(t)|^2 &= \int_{-\infty}^{+\infty} dt x(t) x^*(t) \\&= \int_{-\infty}^{+\infty} dt \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega t} s(\omega) \int_{-\infty}^{+\infty} \frac{d\omega'}{2\pi} e^{-i\omega' t} s^*(\omega') \\&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{d\omega d\omega'}{2\pi} \delta(\omega - \omega') s(\omega) s^*(\omega') \\&= \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} |s(\omega)|^2\end{aligned}$$

■ From Parseval's theorem,

$$\int_{-\infty}^{\infty} x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |s(\omega)|^2 d\omega = E.$$

This implies that power spectrum $|s(\omega)|^2 \Delta\omega$ represents energy concentrated within frequency range of $[\omega, \omega + \Delta\omega]$.

How to define spectrum for stochastic process?

Problem

Fourier transform of stochastic signal $X(t)$ results in different spectra for every trial.

→ Definition necessary for sample average.

Derivation of power spectrum

For finite interval $[-T, T]$, Fourier transform of one realization of stochastic process $X(t)$ is given by

$$X_T(\omega) = \int_{-T}^T X(t) e^{-i\omega t} dt$$

whereas its power is given by $\frac{|X_T(\omega)|^2}{2T} = \frac{1}{2T} \left| \int_{-T}^T X(t) e^{-i\omega t} dt \right|^2$.

Taking the expectation $E\{\cdot\}$, mean power spectra $\bar{X}_T(\omega)$ can be calculated as

$$\begin{aligned} \bar{X}_T(\omega) &= E \left\{ \frac{|X_T(\omega)|^2}{2T} \right\} \\ &= \frac{1}{2T} \int_{-T}^T \int_{-T}^T E\{X(t_1)X^*(t_2)\} e^{-i\omega(t_1-t_2)} dt_1 dt_2 \\ &= \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) e^{-i\omega(t_1-t_2)} dt_1 dt_2. \end{aligned}$$

Supposing $X(t)$ is a wide-sense stationary process, so that $R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$,

$$\bar{X}_T(\omega) = \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{XX}(t_1 - t_2) e^{-i\omega(t_1 - t_2)} dt_1 dt_2$$

Denoting $\tau = t_1 - t_2$,

$$\begin{aligned} \bar{X}_T(\omega) &= \frac{1}{2T} \int_{-2T}^{2T} R_{XX}(\tau) e^{-i\omega\tau} (2T - |\tau|) d\tau \\ &= \int_{-2T}^{2T} R_{XX}(\tau) e^{-i\omega\tau} \left(1 - \frac{|\tau|}{2T}\right) d\tau \geq 0 \end{aligned}$$

Finally, taking the limit of $T \rightarrow \infty$.

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau \geq 0$$

[Wiener-Khinchin Theorem]

- Autocorrelation $R_{XX}(\tau)$ and spectral density $S_{XX}(\omega)$ are related with each other via Fourier transform.

The inverse Fourier transform gives

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega \geq 0$$

In particular, the case of $\tau = 0$ gives the signal power

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega &= R_{XX}(0) \\ &= E\{|X(t)|^2\} \\ &= P \end{aligned}$$

Exercise

Consider a deterministic periodic signal $p(t)$ with a period of $T = 2$ seconds, defined over one period as:

$$p(t) = \begin{cases} 1, & -0.5 \leq t < 0.5 \\ 0, & 0.5 \leq t < 1.5 \end{cases}$$

A stochastic process $X(t) = p(t) + N(t)$, where $N(t)$ is a white Gaussian noise process with a mean of zero and a power spectral density $S_{NN}(\omega) = \sigma^2$.

Suppose you observe a single realization of this stochastic process over a finite time interval $-L \leq t \leq L$. Let this observation be $x_L(t) = p(t) + n(t)$, where $n(t)$ is a sample from the noise process $N(t)$.

- a) Find the Fourier transform $P(j\omega)$ of one period of the deterministic signal $p(t)$.
- b) Describe the expected Fourier transform $X_L(j\omega)$ of the observed signal $x_L(t)$.

Answer

a) $P_1(j\omega) = \frac{2\sin(0.5\omega)}{\omega}$

b) $X_L(j\omega) = F\{p(t)\omega_L(t)\} + F\{n(t)\omega_L(t)\}$, where $\omega_L(t)$ is a rectangular window function of width $2L$ centered at $t = 0$.

Exercise (Cont.)

- What is the autocorrelation function $R_{PP}(\tau)$ of the periodic signal $p(t)$? (Hint: You might find it easier to think about the time average of $p(t)p(t - \tau)$ over one period).
- What is the autocorrelation function $R_{NN}(\tau)$ of the white Gaussian noise $N(t)$ with power spectral density $S_{NN}(\omega) = \sigma^2$? (Recall the inverse Fourier transform relationship between PSD and autocorrelation).

Properties of power spectrum

(1) $S_{XX}(\omega)$ is a real function of ω .

(Because $R_{XX}(-\tau) = R_{XX}^*(\tau)$ and $S_{XX}(\omega) = S_{XX}^*(\omega)$)

(2) $S_{XX}(\omega) \geq 0$.

(3) If $X(t)$ is a real process,

$$\begin{aligned} R_{XX}(\tau) &= E\{X(t + \tau)X(t)\} \\ &= E\{X(s)X(s - \tau)\} = R_{XX}(-\tau). \text{ Therefore} \\ S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau \\ &= \int_{-\infty}^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau + i \int_{-\infty}^{\infty} R_{XX}(\tau) \sin \omega\tau d\tau \\ &= 2 \int_0^{\infty} R_{XX}(\tau) \cos \omega\tau d\tau = S_{XX}(-\omega) \end{aligned}$$

Hence, $S_{XX}(\omega)$ is an even function and can be represented in terms of cos-transform.

Cross-power spectrum

Cross-power spectrum $S_{XY}(\omega)$ of two processes $X(t)$ and $Y(t)$ is

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau,$$

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{i\omega\tau} d\omega,$$

$$S_{YX}(\omega) = S_{XY}^*(\omega).$$

Example of power spectrum

Random variables a_i are mutually uncorrelated, and have mean 0 and variance σ_i^2 . Compute power spectrum of stochastic process:

$$X(t) = \sum_i a_i e^{i\omega_i t}$$

Auto-correlation of $X(t)$ is computed as

$$\begin{aligned} R_{XX}(\tau) &= E\{X(t + \tau)X^*(t)\} \\ &= E\left\{\sum_i a_i e^{i\omega_i(t+\tau)} \sum_k a_k^* e^{-i\omega_k t}\right\} \\ &= \sum_i \sum_k E\{a_i a_k^*\} e^{i(\omega_i - \omega_k)t + i\omega_i \tau} \\ &= \sum_i \sigma_i^2 e^{i\omega_i \tau} \end{aligned}$$

Here, we used $E\{a_i a_k^*\} = 0$ ($i \neq k$), σ_i^2 ($i = k$), due to uncorrelation property of a_i . The power spectrum is

$$\begin{aligned} S_{XX}(\omega) &= \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau \\ &= \sum_i \sigma_i^2 \int_{-\infty}^{\infty} e^{i\omega_i\tau} e^{-i\omega\tau} d\tau \\ &= 2\pi \sum_i \sigma_i^2 \delta(\omega - \omega_i) \end{aligned}$$

Here, we used $\int_{-\infty}^{\infty} e^{i\omega_i\tau} e^{-i\omega\tau} d\tau = 2\pi\delta(\omega - \omega_i)$.

Autocorrelations and the corresponding spectra

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega \iff S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau$$

$\delta(\tau)$	$\iff$	1
1	$\iff$	$2\pi\delta(\omega)$
$e^{j\beta\tau}$	$\iff$	$2\pi\delta(\omega - \beta)$
$\cos \beta\tau$	$\iff$	$\pi\delta(\omega - \beta) + \pi\delta(\omega + \beta)$
$e^{-\alpha \tau }$	$\iff$	$\frac{2\alpha}{\alpha^2 + \omega^2}$
$e^{-\alpha\tau^2}$	$\iff$	$\sqrt{\frac{\pi}{\alpha}} e^{-\omega^2/4\alpha}$

$$e^{-\alpha|\tau|} \cos \beta\tau \quad \Longleftrightarrow \quad \frac{\alpha}{\alpha^2 + (\omega - \beta)^2} + \frac{\alpha}{\alpha^2 + (\omega + \beta)^2}$$

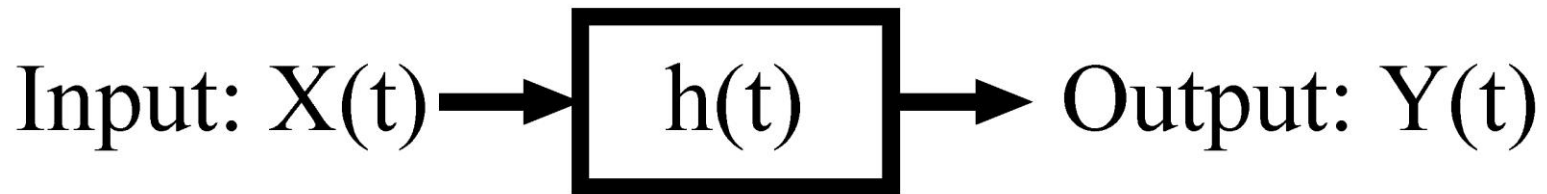
$$e^{-\alpha\tau^2} \cos \beta\tau \quad \Longleftrightarrow \quad \sqrt{\frac{\pi}{\alpha}} \left[e^{-\frac{(\omega - \beta)^2}{4\alpha}} + e^{-\frac{(\omega + \beta)^2}{4\alpha}} \right]$$

$$\begin{cases} 1 - \frac{|\tau|}{T} & |\tau| < T \\ 0 & |\tau| > T \end{cases} \quad \Longleftrightarrow \quad \frac{4\sin^2(\omega T/2)}{T\omega^2}$$

$$\frac{\sin \sigma\tau}{\pi\tau} \quad \Longleftrightarrow \quad \begin{cases} 1 & |\omega| < \sigma \\ 0 & |\omega| > \sigma \end{cases}$$

Power-spectra and Linear System

- Wide sense stationary process $X(t)$ has autocorrelation of $R_{XX}(\tau)$ and its corresponding spectrum $S_{XX}(\omega)$. Suppose $X(t)$ is input to a linear system, whose impulse response is given by $h(t)$. How to compute the output spectrum $S_{YY}(\omega)$?



- From properties of autocorrelation function

$$R_{XY}(\tau) = R_{XX}(\tau) * h^*(-\tau),$$

$$R_{YY}(\tau) = R_{XX}(\tau) * h^*(-\tau) * h(\tau),$$

In general, “If $f(t) \leftrightarrow F(\omega)$ and $g(t) \leftrightarrow G(\omega)$, then, $f(t) * g(t) \leftrightarrow F(\omega)G(\omega)$.” Therefore,

$$\begin{aligned} S_{XY}(\omega) &= F\{R_{XY}(\tau)\} = F\{R_{XX}(\tau) * h^*(-\tau)\} \\ &= S_{XX}(\omega)H^*(\omega) \end{aligned}$$

Here, transfer function is defined as

$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-i\omega t} dt$. The following was used.

$$\begin{aligned} F\{h^*(-\tau)\} &= \int_{-\infty}^{\infty} h^*(-\tau)e^{-i\omega\tau} d\tau \\ &= \left(\int_{-\infty}^{\infty} h(t)e^{-i\omega t} dt \right)^* = H^*(\omega) \end{aligned}$$

Hence,

$$\begin{aligned} S_{YY}(\omega) &= F\{R_{YY}(\tau)\} = S_{XY}(\omega)H(\omega) \\ &= S_{XX}(\omega)|H(\omega)|^2 \end{aligned}$$

■ **Appendix:** “If $f(t) \leftrightarrow F(\omega)$ and $g(t) \leftrightarrow G(\omega)$, then,
 $f(t) * g(t) \leftrightarrow F(\omega)G(\omega)$.”

Proof:

$$\begin{aligned} F\{f(t) * g(t)\} &= \int_{-\infty}^{\infty} f(t) * g(t) e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \right\} e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} f(\tau) e^{-i\omega \tau} d\tau \int_{-\infty}^{\infty} g(t - \tau) e^{-i\omega(t - \tau)} d(t - \tau) \\ &= F(\omega)G(\omega) \end{aligned}$$

($F\{\cdot\}$ represents Fourier transform.)

Example

- Consider white noise with a PSD $S_{XX}(\omega) = \frac{N_0}{2}$ passing through a low-pass filter with a frequency response:

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

- The PSD of the output noise $Y(t)$ will be:

$$S_{YY}(\omega) = S_{XX}(\omega)|H(j\omega)|^2 = \begin{cases} \frac{N_0}{2} \cdot 1^2 = N_0/2, & |\omega| \leq \omega_c \\ \frac{N_0}{2} \cdot 0^2 = 0, & |\omega| > \omega_c \end{cases}$$

- The output noise has a PSD that is band-limited to the cutoff frequency ω_c of the low-pass filter. The total power of the output noise would be the integral of $S_{YY}(\omega)$ over all frequencies.

Applications

- **Filtering in the Frequency Domain:** This fundamental result shows that an LTI system acts as a filter on the power spectrum of the input signal.
- **Frequency Shaping:** The frequency response $H(j\omega)$ determines how different frequency components of the input signal are amplified or attenuated by the system. The power spectrum of the output reflects this shaping.
- **System Identification:** By analyzing the input and output power spectra of a system with a known input (e.g., white noise), we can potentially estimate the magnitude of the system's frequency response $|H(j\omega)|$.